Exposition: How two-layer networks learn in one (giant) gradient step

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Topics in Statistics: Random Matrix Theory in Machine Learning

Abstract

This report engages with **How Two-Layer Neural Networks Learn, One (Giant) Step at a Time** (Dandi et al., 2024) (JMLR, 2024). The reference studies early training of a two-layer network on standard Gaussian inputs when the label follows a **multi-index** model: all low-dimensional structure lies in a link function of projections onto unknown teacher directions. The central question is when a small number of large-batch gradient steps on the first-layer weights produces nontrivial alignment with the teacher subspace $V^\star$—thereby leaving the random-feature (lazy) kernel induced by initialization (Dandi et al., 2024).

Following (Dandi et al., 2024), we separate three ingredients—batch size n, number of gradient steps, and the target's structure (Hermite / leap / staircase)—and stress that they are not interchangeable. In one giant full-batch step, batch scaling $n = \Theta(d)$ supports alignment but only to a single teacher direction unless n reaches $\Theta(d^2)$ scale for multi-direction specialization; still harder directions may demand $n = \Theta(d^\ell)$ with leap index ℓ. With multiple steps and fresh batches of size $n = \Theta(d)$, targets satisfying a staircase coupling can progressively recruit several directions over time, while obstructed pieces retain $\Theta(d^2)$ bottlenecks. Finally, (Dandi et al., 2024) link these geometric facts to approximation and generalization relative to the initialization kernel—including experiments where one gradient step helps or fails versus frozen features.

Critical discussion of assumptions and limitations is in Section 2; the technical overview begins in Section 1.

AI disclosure: Generative AI tools were used for drafting and ideation; the author edited the text and takes responsibility for all content.

1 Setting and contributions

Unless noted, asymptotic statements match (Dandi et al., 2024) (their §1–2; large-d scaling).

1.1 Notation at a glance

Symbol	Meaning
d	Input dimension ($z \in \mathbb{R}^d$).
r	Number of teacher directions.
n	Batch size / samples per update stage.
p	Network width (hidden units).
$W \in \mathbb{R}^{p\times d}$	First-layer matrix; row w_i is neuron-i direction.
$a \in \mathbb{R}^p$	Second-layer vector.
$W^\star$	Teacher matrix (rows $w_j^\star$).
$f^\star, g^\star$	Target: $f^\star(z) = g^\star(W^\star z)$.
$V^\star$	Teacher span $\mathrm{span}\{w_1^\star, ..., w_r^\star\}$.
η	Learning rate for W updates.

ℓ	Leap index (first Hermite order with signal).
$\Theta(\cdot)$	Asymptotic scaling notation.

1.2 Problem setup

Data and target. Covariates $z \propto \mathcal{N}(0, I_d)$ in $\mathbb{R}^d$. Labels follow a multi-index representation: orthogonal teachers $w_1^\star, ..., w_r^\star$ and a link $g^\star : \mathbb{R}^r \to \mathbb{R}$ with

$$y = f^\star(z) := g^\star(\langle w_1^\star, z\rangle, ..., \langle w_r^\star, z\rangle). \tag{1}$$

Isotropic inputs put **all** exploitable structure in $f^\star$. That modeling choice is deliberate: classical kernels depend on isotropic geometry of z but do not automatically exploit hidden low-dimensional structure in y, so improvements over kernel behavior here are evidence of **representation learning** rather than an artifact of input clustering (Dandi et al., 2024).

Two-layer predictor. First- and second-layer parameters $W \in \mathbb{R}^{p\times d}$, $a \in \mathbb{R}^p$, and activation σ,

$$\hat{f}(z; W, a) = \frac{1}{\sqrt{p}} \sum_{i=1}^{p} a_i\, \sigma(\langle w_i, z\rangle). \tag{2}$$

Training regime analyzed. (Dandi et al., 2024) focus on the **early** phase: one or more full-batch gradient steps on W under squared loss, with batch size n coupled to d (typically $n = \Theta(d)$ versus $\Theta(d^2)$ comparisons) and learning-rate scaling tied to width p as in their theorems. For multi-step analyses they use **fresh i.i.d. batches** each iteration and (like related work) treat second-layer training separately—idealizations that simplify the Hermite and concentration analysis (Dandi et al., 2024).

Operational notion of learning. Beating the kernel at initialization means growing overlap between rows of W and $V^\star := \operatorname{span}\{w_1^\star, ..., w_r^\star\}$. Whether overlap arises, and in how many orthogonal directions, is driven jointly by (n, steps) and the Hermite**staircase** structure of $g^\star$ (Dandi et al., 2024).

1.3 Main contributions

1.3.1 A single organizing tension

(Dandi et al., 2024) argue that **specialization**—encoding distinct teacher directions in distinct hidden units—is not a monotone consequence of more data.‘’ Their comparison is between one full-batch update built from n samples (each sample contributes an additive term to the gradient; those terms are independent and can be farmed out to parallel workers, then averaged) and **classical one-pass SGD**, where updates happen one after another so later steps depend on earlier ones. Thus needing a large n for one full batch is not the same as needing many **serial** gradient steps: **sample counts** and **wall-clock complexity** (elapsed real time for the computation to finish, as opposed to an abstract count of serial operations or steps) need not track one another; what matters for their theory is which directions are **algebraic**ally accessible at a given (n, step) pair (Dandi et al., 2024).

1.3.2 One full-batch gradient step on W

Their Theorems 4–5 and Fig. 1 summarize a batch-size hierarchy for one step: alignment already requires n at least on $\Theta(d)$ order with η coupled to p, yet **at most one** direction in $V^\star$ can emerge at that scale when $r > 1$. Acquiring several directions **simultaneously** in one update pushes the batch to $\Theta(d^2)$. Even then, some directions can force $n = \Theta(d^\ell)$ where ℓ is the leap index (Definition 3): the first Hermite degree at which certain components of $g^\star$ appear (Dandi et al., 2024).

1.3.3 Multiple steps: staircase structure and obstructions

Theorem 7 and Fig. 2 describe a different regime: with repeated steps, **fresh** batches of size $n = \Theta(d)$ suffice to learn **multiple** directions over time **provided** new directions are coupled to previous ones through the staircase property (made precise in their §3.4). Intuitively, later directions become statistically predictable once earlier ones align—so the network can climb a sequence of linearized" subproblems. Directions lacking this coupling (and with degenerate low Hermite mass) remain stuck behind the same $\Theta(d^2)$ barrier that already obstructs single-step multi-index learning (Dandi et al., 2024).

1.3.4 Approximation, generalization, and separation from random features

Section 2.3 connects learned subspaces to what degree-k kernels could approximate at initialization (benchmarking against random-features scaling (Mei and Montanari, 2022), as discussed in (Dandi et al., 2024)). Fig. 3-style experiments show when one gradient step on W materially lowers test error relative to random features and when it cannot: the deciding issue is whether the nonlinear signal shares directions the first step can actually align (Dandi et al., 2024).

1.4 Comparison to the reference paper

1.4.1 Ba **et al.** (Ba et al., 2022) as neighbor work

(Ba et al., 2022) study one gradient step on the first layer of a two-layer network in a proportional high-dimensional limit: isotropic Gaussian inputs, single-index teacher, additive noise on y, and a sharp comparison between small versus large learning-rate regimes for representation improvement over the initial kernel (Ba et al., 2022). (Dandi et al., 2024) retain the same early feature learning" spirit but change the target class, the noise model, and the algorithmic questions they prioritize (batch scaling and multiple steps); we spell out overlaps and splits below.

1.4.2 Where the assumptions agree

- **Inputs.** Both works take features to be standard Gaussian (isotropic), so all nontrivial structure is encoded in the label map rather than in anisotropic covariates (Ba et al., 2022; Dandi et al., 2024).
- **Architecture and scaling.** Shallow two-layer networks with the usual $\frac{1}{\sqrt{p}}$ width normalization appear in both analyses (Ba et al., 2022; Dandi et al., 2024).
- **Training protocol (family resemblance).** Both separate learning of the first layer from fitting the second layer and use **fresh** batches for different phases (Ba **et al.** use two independent batches for the two layers in their one-step setup; (Dandi et al., 2024) use a fresh batch at each gradient step when analyzing multi-step training). In either case, decoupling layers avoids coupled finite-width pathologies and matches how this literature idealizes early-time GD (Dandi et al., 2024).
- **Hermite viewpoint.** Each paper expands the target and loss geometry using Hermite analysis on Gaussian inputs; leap-index language enters when discussing which Hermite orders drive alignment (Ba et al., 2022; Dandi et al., 2024).

1.4.3 Where they differ (and what leap index" means in each)

In (Ba et al., 2022) the teacher is **single-index**: $y = f^{\star(x)} + \varepsilon$, with a scalar projection of x driving $f^{\star}$. The comparison table in my presentation slides pins their typical Hermite regime to **no constant Hermite mass**, a nonzero first-order (linear) Hermite contribution, and —in that same schematic— **leap index** equal to 1. Informally, the first nontrivial coupling between target and a teacher direction already appears at the **linear** level, which is the simplest regime for a one-step alignment effect in the proportional limit (Ba et al., 2022).

(Dandi et al., 2024) instead treat **multi-index** teachers $y = g^{\star(W^{\star}z)}$ with r orthonormal directions and **noiseless** labels. Their leap index ℓ is **not** fixed to 1: it indexes how deep one must go in the Hermite expansion along a direction before the target's dependence on that direction turns on." When $\ell = 1$, a toy linear functional $w_1^T z$ has leap one (as in the short leap index toy example" on my presentation slides); when $\ell > 1$, (Dandi et al., 2024) show that even enormous single-batch updates may still fail to learn that direction until n reaches $\Theta(d^\ell)$. Thus Ba **et al.**'s baseline single-index story aligns with the **simplest** leap-1 picture, whereas (Dandi et al., 2024) also treats cases with $\ell > 1$ and with several interacting directions through $g^\star$ (Dandi et al., 2024).

Summary. Similar isotropic-Gaussian + shallow-net setup; dissimilar **target complexity** (single- vs multi-index), **noise** (present vs absent), and **role of ℓ** (Ba **et al.** emphasize a leap-1 single-index tractable case; (Dandi et al., 2024) treat general ℓ and multi-step/ staircase structure).

1.4.4 Map back to (Dandi et al., 2024) (Sections / figures)

Equations (1)–(2) in (Dandi et al., 2024) match our setup block above. Their abstract's three bullets correspond to: §2.1 / Thms. 4–5 / Fig. 1 (single-step batch hierarchy and leap complexity); §2.2 / Thm. 7 / Fig. 2 (multi-step staircase learning at $n = \Theta(d)$); §2.3 / Fig. 3 (approximation and generalization versus initialization kernels, including comparison to random-features scaling (Mei and Montanari, 2022)). Section Section 1.3 tracks that order intentionally (Dandi et al., 2024).

Reading across papers. The progression from (Ba et al., 2022) to (Dandi et al., 2024) is not replace Ba's theorem" but change the task class and ask more detailed resource questions": how large must a **single** batch be to unlock multiple directions in one update, how does leap ℓ obstruct that, and when do **several** large-batch steps at $n = \Theta(d)$ recover staircase targets that a single step cannot (Dandi et al., 2024).

2 Critical discussion and limitations

This section summarizes scope conditions and limitations of (Dandi et al., 2024) and relates them to nearby deep-learning theory.

2.1 Gaussian, noiseless, teacher-only structure

The analysis assumes isotropic Gaussian inputs and **noiseless** labels from a realized multi-index teacher (Dandi et al., 2024). This supports Hermite/Stein/Gaussian-equivalence arguments, but excludes correlated designs, heavy tails, covariate shift, and label noise. (Ba et al., 2022) use additive label noise in their single-index setting, so the two papers compare different statistical regimes. In practice, part of the observed error may be dominated by noise rather than by representation mismatch.

In related statistical and signal-processing settings, controlled noise can act as regularization (noise injection as Tikhonov smoothing) or improve detectability in nonlinear systems (stochastic resonance) (Bishop, 1995; Gammaitoni et al., 1998). Modern deep-learning analyses also study noise scale, batch size, and generalization in SGD-based training (Keskar et al., 2017; Mandt et al., 2017; Smith and Le, 2018).

Can moderate noise ever improve learning dynamics by regularizing alignment, rather than only increasing estimation error?

2.2 Protocol: fresh batches, two stages, symmetrized init

(Dandi et al., 2024) study giant-batch GD on W with fresh samples across steps, then re-estimate the second layer on new data. This two-stage protocol is useful analytically, but differs from common pipelines that reuse minibatches, update (W, a) jointly, and use momentum or adaptive methods. The analysis also uses an auxiliary symmetrized initialization that sets the initial predictor to zero and simplifies the first update. As a result, the reported rates should be interpreted for this training protocol, not for arbitrary optimization settings.

What changes in the early-time feature-learning picture when initialization is not symmetrized?

What changes if $(\boldsymbol{W}, \boldsymbol{a})$ are trained simultaneously with separated time scales (larger step size for the first layer, smaller step size for the second layer), a setup closer to practical training than strict two-stage optimization, and can the paper's conditioning/concentration framework be adapted to analyze this smoothed two-step regime?

Do adaptive optimizers such as Adam induce comparable layerwise time-scale separation in practice (or the opposite, if backpropagated gradients attenuate in earlier layers), and can training be accelerated by schedule designs inspired by this paper that avoid treating all layer parameters identically?

Can the early-time feature-learning phenomena analyzed here motivate alternatives to standard backpropagation (for example, forward-only local objectives such as the forward-forward algorithm (Hinton, 2022)), especially if one seeks biologically plausible update mechanisms?

Generalized-depth alignment question (exact setup): let the teacher be two-layer,

$$y = h^\star(U^\star g^\star(V^\star z)), \tag{3}$$

with $V^\star \in \mathbb{R}^{r_1 \times d}$ (inner teacher directions), $U^\star \in \mathbb{R}^{r_2 \times r_1}$ (outer teacher directions), inner nonlinearity $g^\star : \mathbb{R}^{r_1} \to \mathbb{R}^{r_1}$, and outer map $h^\star : \mathbb{R}^{r_2} \to \mathbb{R}$. Let the student be three-layer,

$$\hat{y} = a^T \sigma_2(W_2 \sigma_1(W_1 z)), \tag{4}$$

with $W_1 \in \mathbb{R}^{p_1 \times d}$ and $W_2 \in \mathbb{R}^{p_2 \times p_1}$. Under what conditions do we get staged alignment $\mathrm{Row}(W_1^t) \to \mathrm{Row}(V^\star)$ and $\mathrm{Row}(W_2^t) \to \mathrm{Row}(U^\star)$ during early training? Which geometric conditions (principal-angle / range-nullspace separation between teacher and student subspaces) and which complexity descriptors of $g^\star, h^\star$ (e.g., separate leap indices, Hermite tensor ranks, derivative bounds) are sufficient for this behavior?

2.3 Preliminary experiments addressing open questions

To partially address the open questions above (label noise and symmetrized initialization), I ran paper-style one-step experiments with Gaussian inputs and explicit teacher functions $f^{\star(z)}$, while varying (i) initialization protocol and (ii) additive label-noise level.

Protocol used in this report's experiments. I compare Eq. (10) i.i.d. initialization to Eq. (10)+(11) symmetrized initialization, and clean labels ($\sigma = 0$) to noisy labels $y = f^{\star(z)} + \varepsilon$ with $\varepsilon \propto \mathcal{N}(0, \sigma^2)$ on a sweep $\sigma \in \{0, 0.05, 0.1, 0.2, 0.4\}$. In each condition, I compare frozen-W to one giant step on W, then ridge-fit the readout layer.

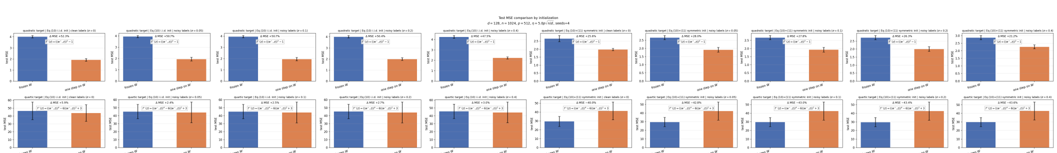

Figure 1: Preliminary one-step experiment (single-index targets): test MSE comparison across Eq. (10) i.i.d. vs Eq. (10)+(11) symmetric initialization, and across noise levels $\sigma \in \{0, 0.05, 0.1, 0.2, 0.4\}$. Each panel compares frozen-W and one-step-W.

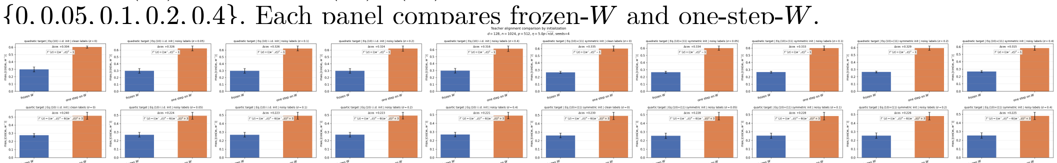

Figure 2: Same sweep as above, but reporting max row-teacher alignment. One-step updates continue to increase alignment under moderate noise, though this does not automatically imply lower test error in every target/init condition.

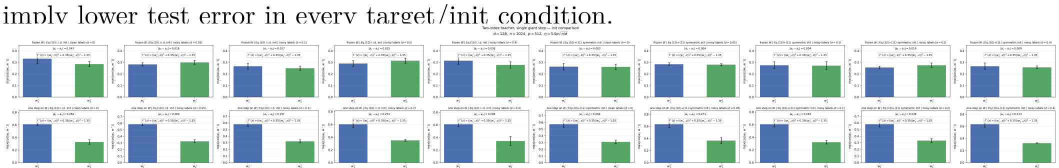

Figure 3: Two-index preliminary experiment: $f^{\star(z)} = (\langle w_1^\star, z\rangle)^2 + 0.35(\langle w_2^\star, z\rangle)^2 - 1.35$. Panels compare frozen-vs-one-step alignment, split by initialization and noise level. In this setup, one-step training still exhibits directional concentration, while added noise mainly shifts error levels rather than producing a clear stochastic-resonance gain.

Preliminary takeaway (not a theorem). In these finite-width runs, I do not observe a robust moderate-noise improves test error’‘ effect. The dominant pattern is that one-step updates still increase alignment under noise, while test error typically worsens or remains similar as σ grows. Symmetrized initialization changes the baseline predictor and can materially alter apparent one-step gains, consistent with the paper's caution that protocol details matter for interpreting early-time feature learning.

2.4 Target geometry: regularity, staircase, leap, obstructed targets

The main bounds use staircase/polynomial structure in $g^\star$ and leap index ℓ along each direction (Dandi et al., 2024). Figure 2 includes cases where no teacher direction is recovered in finite depth, and some directions require batch scaling $\Theta(d^\ell)$. These are algebraic constraints in Hermite space (which orders carry mass along which directions). This perspective is different from landscape-based analyses of non-convex optimization and does not directly cover deeper compositional architectures.

Do recently proposed non-backprop / random-search post-training approaches (e.g., random perturbation experts in (Gan and Isola, 2026) and hyperscale ES variants in (Sarkar et al., 2026)) admit a connection to staircase-style sequential feature recruitment, or do they rely on a different geometric mechanism?

Related jumping staircase’‘ question: if recovery of a new direction at Hermite order $k+1$ depends on how many tensor factors are already shared with learned order-k structure, does the theory imply an intermediate regime where $n = \Theta(d^2)$ (rather than $\Theta(d)$) can unlock additional directions in one giant step when overlap is partial? If such jumps exist, can they motivate layerwise learning-rate or batch-size schedules that differ across layers instead of uniform adaptive updates?

2.5 RMT and concentration: n of order d and the Gaussian toolbox

Operator-norm concentration for empirical covariances already requires n to scale with d when directions are unconstrained. This underlies the paper's batch-size thresholds before leap/staircase refinements. The proof techniques (projection conditioning, Gaussian equivalence, Hermite calculus) are specialized to Gaussian covariates, so extension to non-Gaussian designs requires new arguments.

2.6 What is proved vs conjectured (Table 1) + scope in DL theory

Table 1 in (Dandi et al., 2024) includes both proved results and entries labeled as educated guesses, so claims should be distinguished accordingly. Relative to broader theory, the paper focuses on finite-time, finite-width directional alignment in $V^\star$, while NTK/lazy and mean-field lines emphasize different asymptotic objects. The structural contribution is the interaction of batch size, iteration depth, and Hermite geometry of $g^\star$; conclusions that depend on Gaussian noiseless inputs, fresh batches, and decoupled training should be read as model-specific.

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A Appendix

Optional: longer calculations, extra figures, or extended experiments.